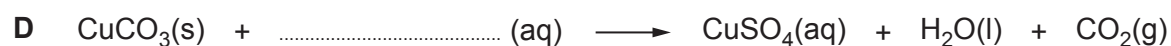


3. (a) **A, B, C** and **D** are equations for reactions used to make salts.

(i) Complete the equation for each reaction.

[4]



(ii) Outline the **three** steps you would carry out to get a **pure** sample of silver chloride, AgCl, from the reaction mixture in equation **B**.

[2]

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- (b) Sodium chloride is made when sodium hydroxide reacts with hydrochloric acid.

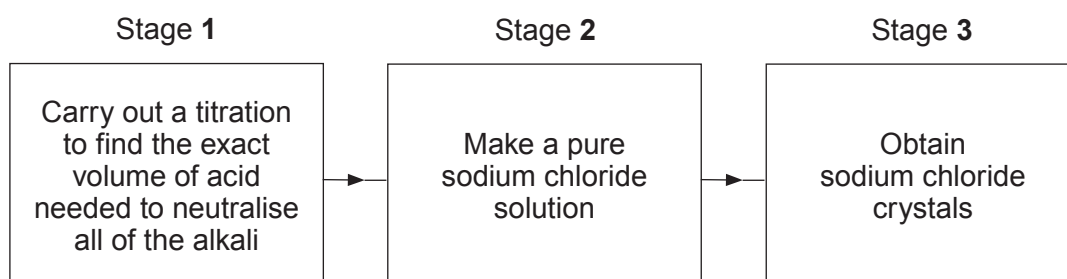


- (i) Write a balanced equation for the reaction.

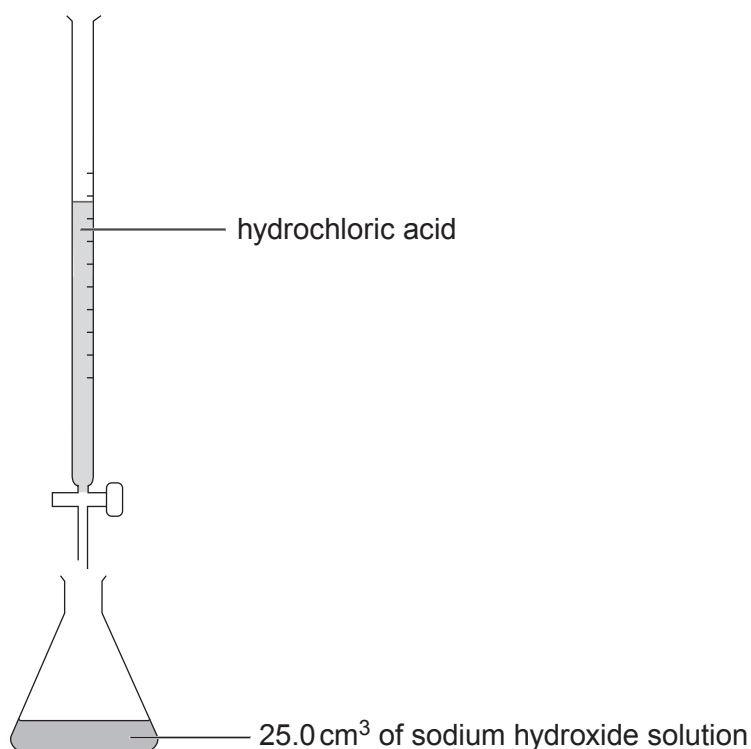
[2]

- (ii) A student was asked to prepare pure sodium chloride crystals using 25.0 cm³ of sodium hydroxide solution.

The flow diagram shows the student's plan.



I. The student sets up the apparatus below to carry out stage 1.



Describe how the student will find the exact volume of acid needed to neutralise all of the alkali.

[2]

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II. The neutral solution obtained in stage 1 is not pure.

Describe what the student must do in stage 2 to make a pure sodium chloride solution.

[2]

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III. State what the student must do in stage 3 to obtain sodium chloride crystals.

[1]

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